

04

COST-CUTTING IN AI SPENDS

While we all absolutely love using GenAI tools like ChatGPT, Gemini, Copilot and others, let me tell you that training and operating large-scale generative AI models, such as those developed by OpenAI, Google, and Anthropic, are an expensive business and not everyone's cup of tea. For instance, OpenAI's GPT-4 model, released in 2023, incurred an estimated training cost of \$78.4 million. Beyond initial training, the operational costs of deploying these models are considerable as well. For example, inference – the process of running trained models to generate outputs – requires substantial computational resources, leading to constant ongoing expenses. For example, Dario Amodei, CEO of Anthropic, noted that training AI models can cost around \$100 million at the lower end, with some models in development reaching up to a billion dollars.



This doesn't take into account the cost of AI hardware, which can truly make your eyes water. Just take a look at NVIDIA's Blackwell components' costs. The GB200 Grace Blackwell superchips can be as much \$70,000 each, while the Blackwell B100 accelerator isn't exactly cheap at \$30,000 to \$35,000. Server racks with multiple of these AI workload chips running together can easily go over a million dollars, if not more. The Blackwell chips are expected to be available for volume sales in early 2025. These fig-

ures highlight the financial challenges faced by organisations in sustaining and scaling generative AI technologies.

The cost of training these large AI models has been a significant concern for AI companies – however, several developments suggest that these costs are expected to decrease by 2025. Companies like Amazon Web Services (AWS) are investing in custom AI chips, such as Trainium, to offer more cost-effective alternatives to traditional GPUs. AWS announced plans to provide \$110 million in free computing credits to AI researchers, aiming to challenge NVIDIA's dominance and reduce training expenses. The proliferation of open-source AI models like Meta's Llama allows organisations to leverage pre-trained models, significantly cutting down on the resources required for training from scratch. This approach democratises access to advanced AI capabilities and reduces associated costs. Innovations in training methodologies, such as model distillation and transfer learning, enable the development of smaller, more efficient models that require less computational power and time to train, thereby lowering costs.

Ransomware has evolved significantly through 2024, adopting more sophisticated and multifaceted tactics. Cybercriminals have increasingly leveraged AI to enhance the speed and precision of their attacks, enabling rapid propagation across networks and complicating early detection efforts. The rise of Ransomware-as-a-Service (RaaS) models has democratised access to advanced ransomware tools, allowing even less experienced actors to launch sophisticated attacks – a notable example is the attack on Newpark Resources, a major oilfield supplier, in November 2024, which disrupted critical systems and highlighted the accessibility of ransomware tools to a broader range of cybercriminals. The adoption of double and triple extortion

05

NO RESPITE FROM RANSOMWARE



tactics has become more prevalent. In October 2024, the University of Paris-Saclay experienced such an attack, where the RansomHouse group threat-

ened to disclose 1 terabyte of sensitive data, including personal documents, unless a ransom was paid.

Furthermore, as quantum computing advances, ransomware groups may begin employing quantum-proof encryption techniques, making it nearly impossible for victims to decrypt their data without paying the ransom. Critical infrastructure sectors, such as energy, water, and healthcare, have become prime targets, with attackers seeking maximum impact and leverage by disrupting essential services.